

Tutorial No. 3

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Problem Statement : Identify and classify functional and non-functional requirements for a hospital management system based on given scenarios.

Solution :

Definition of Functional Requirement

A functional requirement specifies what a system must do. It describes the functions, services, or behaviours the system should provide to fulfill user needs.

Definition : Functional requirements define the specific operations and functions that a system must perform.

Definition of Non-functional requirement

A non-functional requirement specifies how a system should perform. It defines the quality attributes, performance standards, and constraints of the system.

Definition : Non-functional requirements describe the quality, performance, and limitations of a system rather than its functions.

Hospital Management System (HMS)

Functional Requirements (FR)

FR define what the system should do.

Patient registration - The system shall allow staff to register new patients and maintain patient records.

Appointment Scheduling - The system shall enable patients to book, modify, or cancel doctor appointments.

Doctor management - The system shall store and manage doctor details, specializations, and availability.

Billing and payments - The system shall generate bills and record payments for consultations, tests, and treatments.

Medical Records management - The system shall store patient medical history, prescriptions, and lab reports.

Laboratory management - The system shall manage test records and record lab results.

User authentication - The system shall allow login for different users (admin, doctor, receptionist).

Non-functional requirements (NFR)

~~Non~~-functional requirements define how well the system should perform.

Performance - The system should respond to user requests within 2-3 seconds.

Security - The system should ensure secure data to patient data using authentication and

authorization

- Reliability - The system should be available 24/7 with minimal downtime.
- Usability - The system should be easy to use for hospital staff with minimal training.
- Scalability - The system should support an increasing number of users and patient records.
- Maintainability - The system should allow easy updates and bug fixes.
- Data backup and recovery - The system should regularly back up data and recover it in case of failure.

Classification Summary

Type	Description	Examples
Functional	Services provided by system	Registration, billing, appointment
Non-Functional	Quality attributes of system	Security, Performance, usability

Conclusion:

- Functional requirements describe system functionalities, while non-functional requirements focus on quality, performance, and constraints of the hospital management system.